BUSFIN 8200: Asset Pricing Foundations
Problem Set 2

October 22, 2026

Question 1

Question 2

2a)

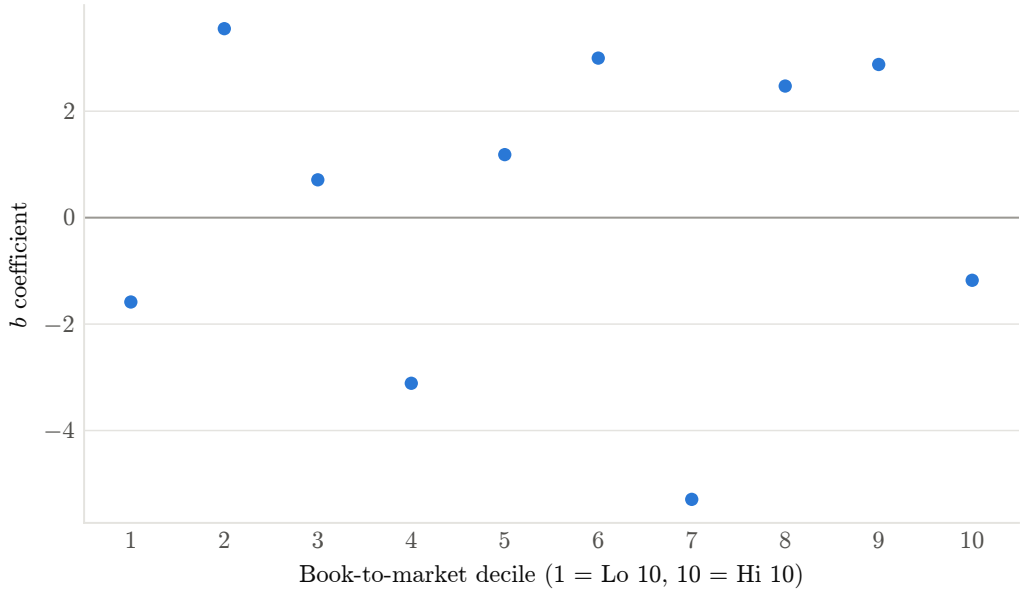


Figure 1: Coefficients b of the SDF $M_t^* = a - b'xR_t$ that prices the monthly excess returns of the 10 value-weighted book-to-market decile portfolios (July 1927 to August 2026), under the normalization $\widehat{\mathbb{E}}[M^*] = 1$. Excess returns are in decimal units.

2b)

The sample (Pearson) correlation between the monthly SDF series $M^*_{\text{BM},t}$ estimated in 2a) and $M^*_{\text{MOM},t}$ estimated in 2b), over July 1927 to August 2026 ($T = 1190$ months), is

$$\widehat{\text{Corr}}\left(M^*_{\text{BM},t},\, M^*_{\text{MOM},t}\right) = 0.355.$$

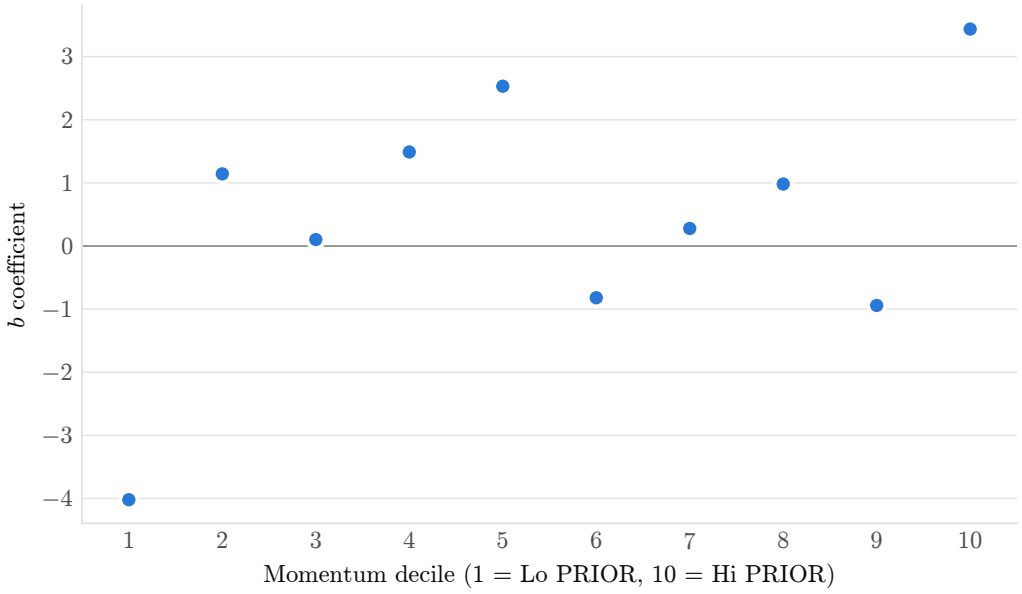


Figure 2: Coefficients b of the SDF $M_t^* = a - b' x R_t$ that prices the monthly excess returns of the 10 value-weighted momentum (prior 12–2 month return) decile portfolios (July 1927 to August 2026), under the normalization $\widehat{\mathbb{E}}[M^*] = 1$. Excess returns are in decimal units.

2c)

Table 1: Monthly Sharpe ratios $\widehat{SR}(xR^*) = \widehat{\mathbb{E}}[xR_t^*]/\widehat{\sigma}(xR_t^*)$ of the portfolios $xR_t^* = b' x R_t$ implied by the SDFs of 2a)–2c), July 1927 to August 2026 ($T = 1190$ months). Not annualized.

SDF	Test assets	$\widehat{SR}(xR^*)$
2a)	10 book-to-market deciles	0.192
2b)	10 momentum deciles	0.258
2c)	20 book-to-market and momentum deciles	0.305

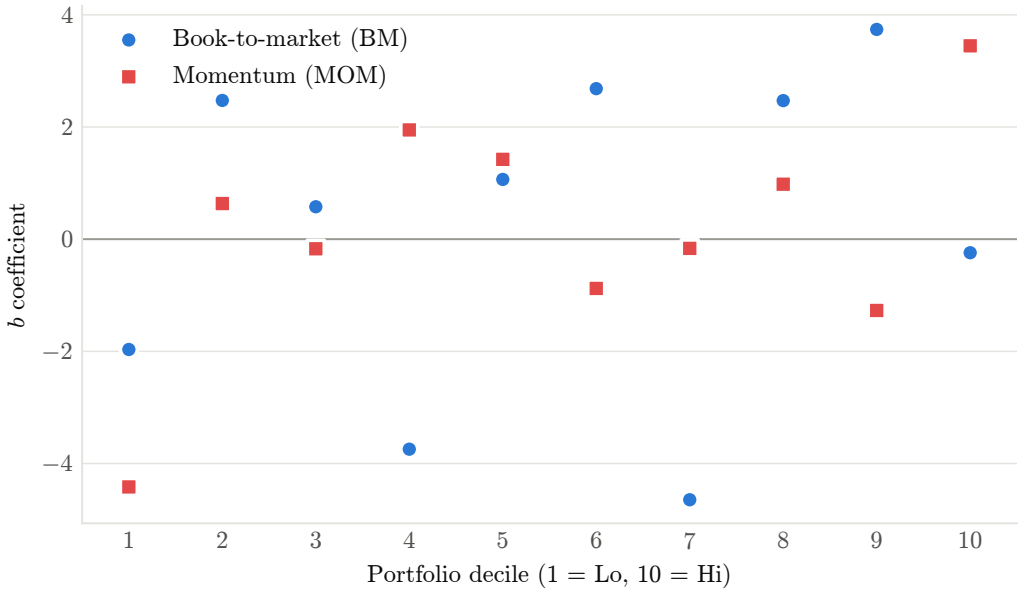


Figure 3: Coefficients b of the SDF $M_t^* = a - b'xR_t$ that prices the monthly excess returns of the 20 value-weighted portfolios in the merge of the book-to-market (blue circles) and momentum (red squares) decile portfolios (July 1927 to August 2026), under the normalization $\widehat{\mathbb{E}}[M^*] = 1$. Excess returns are in decimal units.

2d)

[Explanation will be here]

	SDF M_t^* estimated in		
Portfolio	2a) BM	2b) MOM	2c) BM + MOM
Panel A: book-to-market deciles			
1 (Lo 10)	0.00	−0.15	0.00
2	0.00	−0.01	0.00
3	0.00	−0.02	0.00
4	0.00	−0.08	0.00
5	0.00	0.05	0.00
6	0.00	0.13	0.00
7	0.00	0.03	0.00
8	0.00	0.24	0.00
9	0.00	0.34	0.00
10 (Hi 10)	0.00	0.32	0.00
Panel B: momentum deciles			
1 (Lo PRIOR)	−1.02	0.00	0.00
2	−0.51	0.00	0.00
3	−0.33	0.00	0.00
4	−0.15	0.00	0.00
5	−0.09	0.00	0.00
6	−0.03	0.00	0.00
7	0.05	0.00	0.00
8	0.19	0.00	0.00
9	0.22	0.00	0.00
10 (Hi PRIOR)	0.54	0.00	0.00

Table 2: Pricing errors $\hat{\alpha}_p = \widehat{\mathbb{E}}[xR_p] - \widehat{\mathrm{Cov}}[-M^*, xR_p]$ of the 20 book-to-market and momentum decile portfolios under the SDFs M_t^* estimated in 2a) (10 book-to-market deciles), 2b) (10 momentum deciles) and 2c) (all 20 portfolios), in percent per month. Monthly excess returns, July 1927 to August 2026 ($T = 1190$ months); covariances divide by T.

2e)

[Explanation will be here]

Portfolio	$\hat{\alpha}_{OS,p}$
Panel A: book-to-market deciles	
1 (Lo 10)	1.10
2	0.79
3	0.90
4	0.70
5	0.80
6	0.72
7	0.70
8	0.59
9	0.83
10 (Hi 10)	0.97
Panel B: momentum deciles	
1 (Lo PRIOR)	2.71
2	1.65
3	1.39
4	1.21
5	0.92
6	0.83
7	0.76
8	0.64
9	0.43
10 (Hi PRIOR)	0.55

Table 3: Out-of-sample pricing errors $\hat{\alpha}_{OS,p} = \widehat{\mathbb{E}}[xR_p] - \widehat{\mathrm{Cov}}[-M^*_{OS}, xR_p]$ of the 20 book-to-market and momentum decile portfolios, in percent per month. $\hat{a}$ and $\hat{b}$ are estimated as in 2c) on July 1927 to December 1989 ($T = 750$ months); $M^*_{OS,t} = \hat{a} - \hat{b}' xR_t$ and the moments are computed on January 1990 to August 2026 ($T = 440$ months); covariances divide by T.

Question 3

Question 4